Tapeout Review: Digital CIM

Yunqi's Children

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Chip Overview

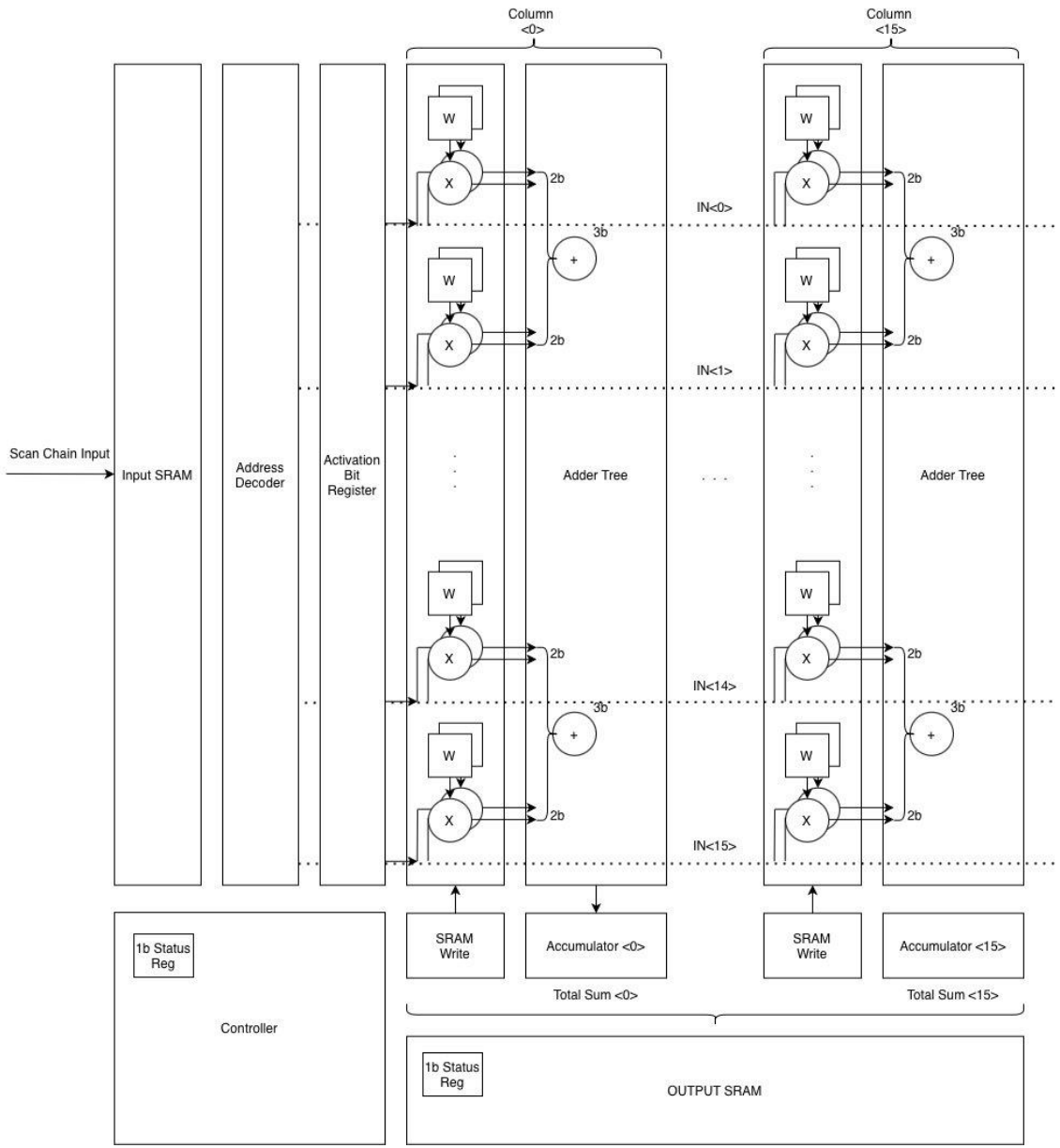
Final Design:

- 16x16 Digital CIM core
  - 100% synthesizable RTL using level-sensitive latches
- Clock Frequency: 500 MHz
- 1b weights and 4b activations
- 1 mode of operation
  - Start computation

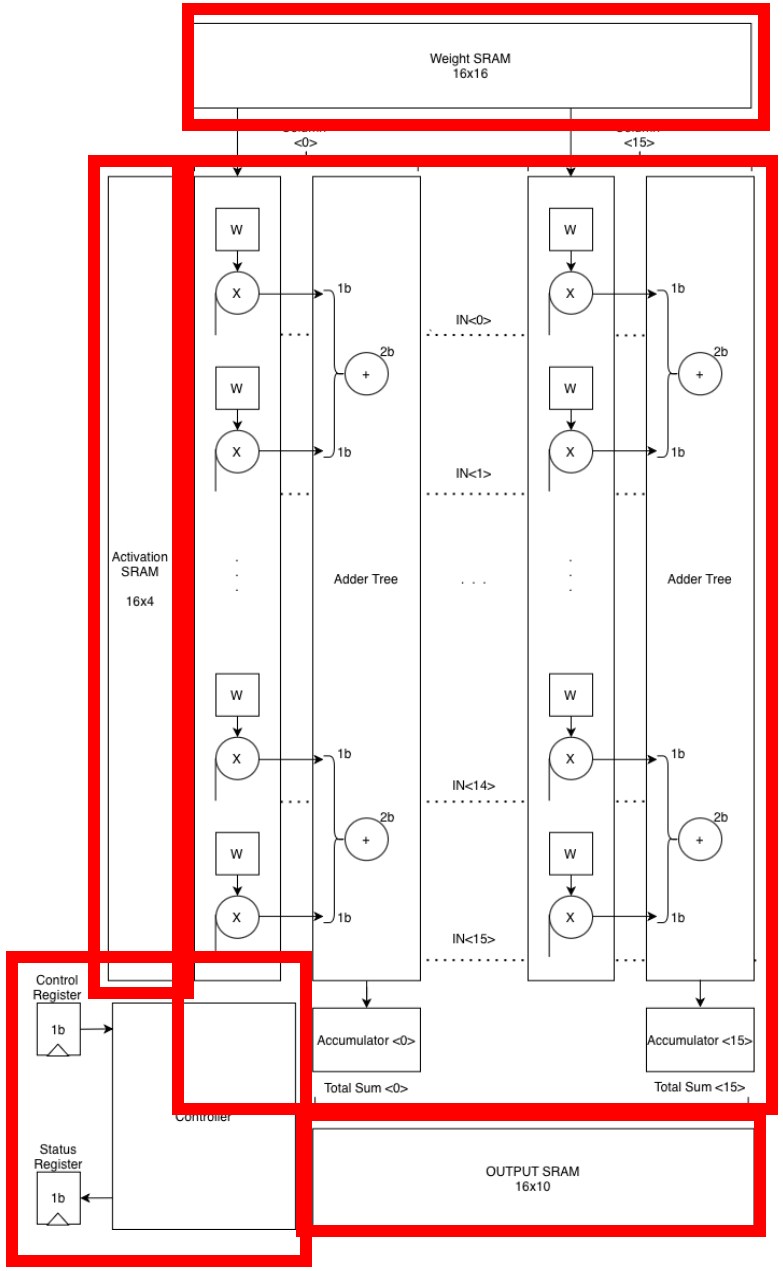
Chip Overview

	Architecture Review	Detailed Design Review	Physical Design Review
2-bit weights	2b weights 4b activations	1b weights 4b activations	1b weights 4b activations
Clock	50 MHz	500 MHz	500 MHz
SRAM	Single-port SRAM (Memory Compiler)	Register (Synthesizable register array)	Register (Synthesizable register array)
Implementation	Hand layout	Hand layout	Synthesizable RTL

OLD


Column <0>
Column <15>
Scan Chain Input
Input SRAM
Address Decoder
Activation Bit Register
W
X
2b
3b
+
IN<0>
IN<1>
IN<14>
IN<15>
Adder Tree
SRAM Write
Accumulator <0>
Accumulator <15>
Total Sum <0>
Total Sum <15>
1b Status Reg
Controller
OUTPUT SRAM

NEW


Weight SRAM 16x16
Activation SRAM 16x4
W
X
1b
2b
+
IN<0>
IN<1>
IN<14>
IN<15>
Adder Tree
Accumulator <0>
Accumulator <15>
Total Sum <0>
Total Sum <15>
Control Register
Status Register
Controller
OUTPUT SRAM 16x10

Timing

TT Corner Analysis with .200 uncertainty:

Hold Time Slack: .205

Path: Launched from controller

Setup Time Slack: .752

Critical Path: Activation bits SRAM -> Accumulator Register

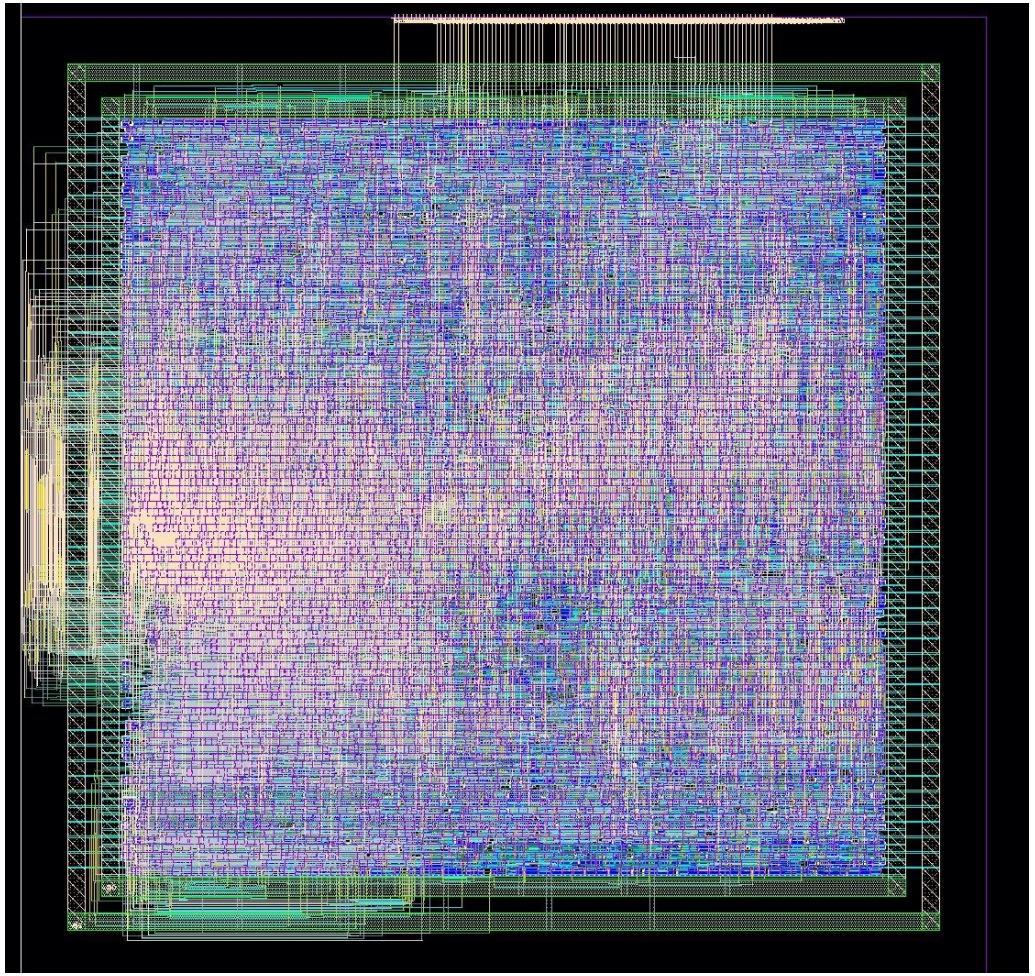
```
Path 1: MET Hold Check with Pin core_cim_wt_data_q_reg_7_0/CP
Endpoint:   core_cim_wt_data_q_reg_7_0/D              (v) checked with  leading
edge of 'clk'
Beginpoint: u_digital_cim/u_controller/state_reg_2_/Q (^) triggered by  leading
edge of 'clk'
Path Groups: {clk}
Analysis View: typical_analysis_view
Other End Arrival Time          0.015
+ Hold                          0.033
+ Phase Shift                   0.000
- CPPR Adjustment               0.000
+ Uncertainty                   0.200
= Required Time                 0.248
  Arrival Time                  0.453
  Slack Time                    0.205
     Clock Rise Edge                 0.000
     + Clock Network Latency (Prop)  0.157
     = Beginpoint Arrival Time       0.157
Path 1: MET Setup Check with Pin u_digital_cim/say_less_9__u_accumulator/acc_
reg_reg_9_/CP
Endpoint:   u_digital_cim/say_less_9__u_accumulator/acc_reg_reg_9_/D (^)
checked with  leading edge of 'clk'
Beginpoint: u_act_sram/rdata_reg_1_/Q                                (^)
triggered by  leading edge of 'clk'
Path Groups: {clk}
Analysis View: typical_analysis_view
Other End Arrival Time          0.157
- Setup                         0.008
+ Phase Shift                   2.000
+ CPPR Adjustment               0.000
- Uncertainty                   0.200
= Required Time                 1.950
- Arrival Time                  1.198
= Slack Time                    0.752
     Clock Rise Edge                 0.000
     + Clock Network Latency (Prop)  0.015
     = Beginpoint Arrival Time       0.015
```

Layout and Integration

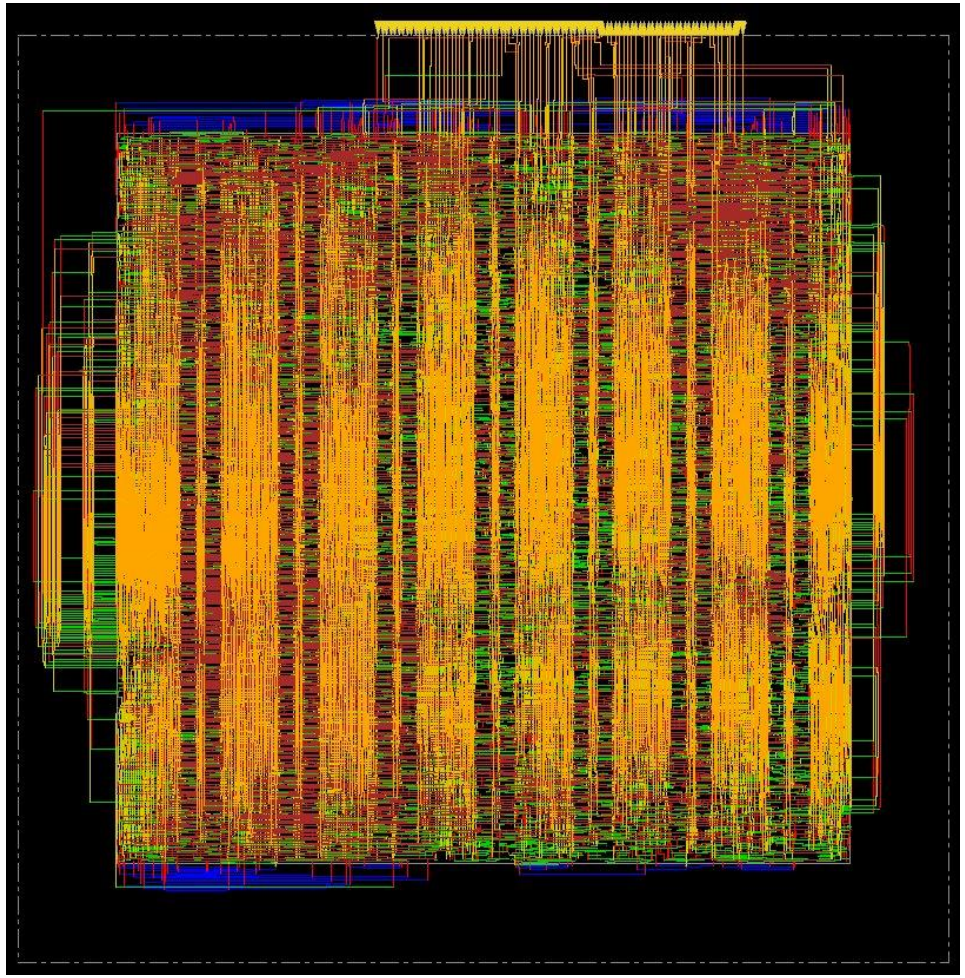
- Design dimensions, utilization – 51
- IO pads assignments:

```
logic rst_n, bypass_i, clk_i;
logic scan_phi, scan_phi_bar, scan_data_in, scan_load_chain, scan_load_chip, scan_id;
wire  scan_data_out, slow_clk_o, halt;
```

Full Design Layout & Signal Routing



Full Design Layout



Signal Routing – 100%

Timing Signoff

- Mode/corner - TT
- Setup: slack, #violating endpoints: **None**

```
------------------------------------------------------------------
     optDesign Final SI Timing Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.752  |  0.752  |  1.005  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   852   |   820   |   76    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |     11 (11)      |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 93.134%
------------------------------------------------------------------
```

Timing Signoff

- Hold: slack, #violating endpoints: **None**

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.205  |  0.205  |  0.289  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   852   |   820   |   76    |
+--------------------+---------+---------+---------+

Density: 93.134%
------------------------------------------------------------------
```

DRC Signoff

- Summary: 0 errors (or list waivers with justification)
- "DRC Clean": All waivable errors

- Top categories previously failing and how resolved
- Screenshot(s) or table of final clean results

Check / Cell	Results	Flat
Check DOD.R.1	1	1
Check PO.DN.2	2	2
Check PO.R.8	136	4227
Check DPO.R.1	1	1
Check M5.DN.1	1	1
Check M6.DN.1	2	2
Check M7.DN.1	1	1
Check M8.DN.1	1	1
Check M9.DN.1:L	1	1
Check M9.DN.2:L	1	1
Check CSR.R.1:NWi	42	42
Check CSR.R.1:PPi	122	122
Check CSR.R.1:NPi	122	122
Check CSR.R.1:COi	1000	1000
Check CSR.R.1:M1i	1000	1000
Check CSR.R.1:M1_real	1000	1000
Check CSR.R.1:M2i	1000	1000
Check CSR.R.1:M2_real	1000	1000
Check CSR.R.1:M3i	1000	1000
Check CSR.R.1:M3_real	1000	1000
Check CSR.R.1:M4i	573	573
Check CSR.R.1:M4_real	573	573
Check CSR.R.1:M5i	312	312
Check CSR.R.1:M5_real	312	312
Check CSR.R.1:M6i	127	127
Check CSR.R.1:M6_real	127	127
Check CSR.R.1:M7i	8	8
Check CSR.R.1:M7_real	8	8
Check CSR.R.1:VIA1i	1000	1000
Check CSR.R.1:VIA2i	1000	1000
Check CSR.R.1:VIA3i	1000	1000
Check CSR.R.1:VIA4i	1000	1000
Check CSR.R.1:VIA5i	1000	1000
Check CSR.R.1:VIA6i	1000	1000
Check CSR.R.1:ODi	1000	1000
Check CSR.R.1:POi	1000	1000
Check DM1.R.1	1	1
Check DM2.R.1	1	1
Check DM3.R.1	1	1
Check DM4.R.1	1	1
Check DM5.R.1	1	1
Check DM6.R.1	1	1
Check DM7.R.1	1	1
Check DM8.R.1	1	1
Check DM9.R.1	1	1
Check ESD.WARN.1	1	1

DRC Signoff

Top categories previously failing and how resolved

- Failing hold-time violations during initial APR
  - Resolved by identifying false paths from scan chain to the FF's
- Failed top-level integration with other due to name dependencies in apr.v
  - Resolved by changing the individual module names in apr.v files that had module dependencies

LVS Signoff

- Top-level LVS result: clean
- Warning: Unbalanced smashed MOSFETS were matched

chip_top.lvs.report

```
                ##################################################
                ##                                              ##
                ##        C A L I B R E   S Y S T E M           ##
                ##                                              ##
                ##            L V S   R E P O R T               ##
                ##                                              ##
                ##################################################



REPORT FILE NAME:         chip_top.lvs.report
LAYOUT NAME:              svdb/chip_top.sp ('chip_top')
SOURCE NAME:              /afs/umich.edu/group/e/ece598to/w26/group1/digital_cim/v2lvs/chip_top.cdl ('chip_top')
RULE FILE:                _calibre.lvs_
CREATION TIME:            Tue Mar 17 16:07:28 2026
CURRENT DIRECTORY:        /afs/umich.edu/group/e/ece598to/w26/group1/digital_cim/final/Calibre/LVS
USER NAME:                paaditya
CALIBRE VERSION:          v2025.3_28.17    Fri Aug 1 15:10:27 PDT 2025



                             OVERALL COMPARISON RESULTS



                       #      ##################      _   _
                      #       #                #     *   *
                  #   #       #     CORRECT    #       |
                   # #        #                #     \___/
                    #         ##################

  Warning:  Unbalanced smashed mosfets were matched.
  Warning:  Ambiguity points were found and resolved arbitrarily.

*****************************************************************************************************
                                        CELL  SUMMARY
*****************************************************************************************************

  Result         Layout                       Source
  -----------    -----------                  --------------
  CORRECT        chip_top                     chip_top



*****************************************************************************************************
                                        LVS PARAMETERS
*****************************************************************************************************
```

ERC / Connectivity etc.

- 0 ERC violations detected → Clean connectivity
- All power/ground, well, and substrate connections are valid
- Some runtime warnings (unmapped layers, missing net names) exist but do not produce ERC errors
- Design is ERC clean

DFT

- Test clocks/resets, scan enable routing
  - Scan in activations and weights
  - Scan in 1 to control register to start core
  - Status register outputs 1 when the core is done with computation
- Any test-mode timing closure summary
  - Post-APR simulations ran on minimum delay for hold-time violations and maximum delay for setup time violations, getting a frequency of 500 MHz

DFT

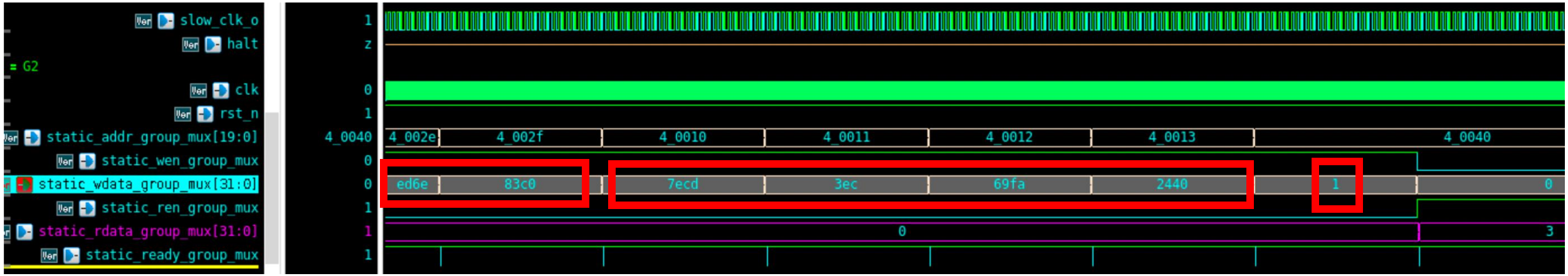
- Scan in weights and activations
- Scan in 1 to start core

group1 > digital_cim > vsim > tests > test0_activations.hex

```
7ECD
03EC
69FA
2440
```

group1 > digital_cim > vsim > tests > test0_weights.hex

```
6656
9D8A
D11A
0DF9
7AF9
3796
28CC
A509
B44E
5C7C
313F
BB3C
035D
D233
ED6E
83C0
```

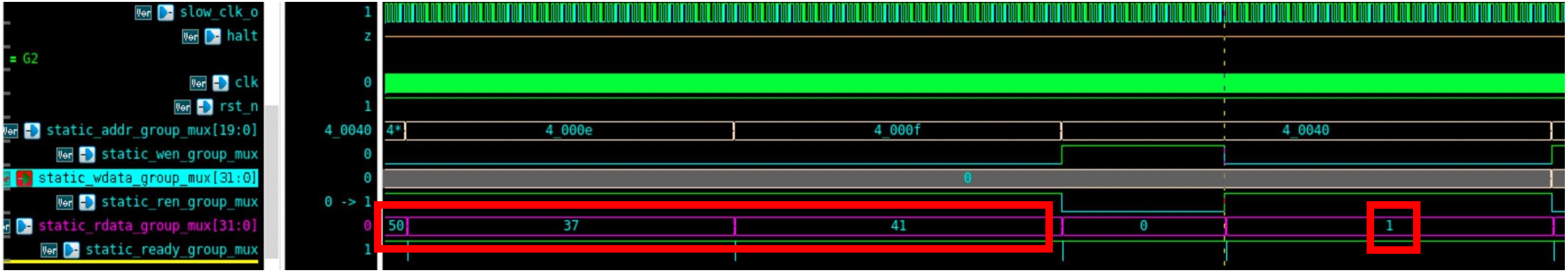


DFT

- Scan out computed results
- Once completed, status register flips to 1

group1 > digital_cim > vsim > tests > test0_expected_mac.hex

```
003A
0040
0054
0072
005C
0044
004B
0027
0055
002D
0048
0041
0046
0050
0037
0041
```



Final PPA Results

- Total area = **51,437.880 um^2**
- Core area: **31969.080 um^2**
- Chip density(Counting Std Cells and MACROS: **62.151%**
  - Effective Utilization: 100%
- Power: 19.37 mW under TT
  - Total Internal: 13.0352 mW
  - Total Switching: 6.327 mW
  - Total Leakage: 0.01224 mW

Open Issues, Waivers, and Risk Assessment

Issue	Severity	Owner	Mitigation	Status
Scan-Chain	Not severe	Aditya	Robust testing	Correct
Core	Not severe	Minjae	Robust testing with various test vectors	Correct
Controller	Not severe	Harshit	Integration and testing with core	Correct
Integration	Not severe	Minjae	Tested block with I/O pads and internal clock	Correct

- Any remaining waivers - None
- Schedule risks and contingency plan